Uniwersytet SWPS

Task list no. 4 – We want to work with stacks, queues, and doubly linked lists.

Task 1. Implement a queue that supports the following operations:

1. Enqueue (push): insert an element at the rear of the queue.
2. Dequeue (pop): remove an element from the front of the queue.
3. Display: Print all elements currently in the queue

Task 2. Implement a doubly linked list with structure in C that supports the following operations:

1. Insert a new node at the beginning.
2. Insert a new node at the end.
3. Insert a new node after a specified node.
4. Delete a node from a list by key
5. Find a node with a specified key.

Task 3. There are **6 students** (labeled 1 to 6). Friendships are as follows:

1. Student 1 is friends with 3
2. Student 3 is friends with 4
3. Student 1 is friends with 2
4. Student 5 is friends with 6

Write a C program using Disjoint Sets to represent the friendship groups and print them.

Task 4. Please test your code for Task 3 using a random friendship network of size 100.

The lab report should contain your code along with the results.

Note. ALL PROGRAMS SHOULD COMPILE WITHOUT ERROR AND WORK (i.e. DO EXACTLY WHAT IS DESCRIBED IN THE TASK)!!!

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